

“FLIGHT CENTER” Reservation System Documentation

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1. Introduction

This document provides detailed information about the airline reservation system project. It describes the system's functionality, objectives, and technical aspects.

2. Objectives

- To develop an automated system for airline reservations.
- To streamline the booking and cancellation process.
- To provide an efficient and user-friendly interface for customers.

3. Scope

The system is designed for Flight Center to manage bookings, flight schedules, customer data, and payments. It supports both online and offline modes.

4. System Features

- User account management (registration/login).
- Flight search and availability check.
- Seat selection and reservation.
- Online payment and invoice generation.
- Admin dashboard for flight and schedule management.

5. System Requirements

Hardware Requirements

Processor: Intel Core i3 or higher
RAM: 4 GB or higher
Storage: 50 GB minimum

Software Requirement

Operating System: Windows/Linux/Mac
Backend: MySQL, PostgreSQL, or MongoDB
Frontend: HTML, CSS, JavaScript, React/Angular
Server: Node.js, Python (Django/Flask), or PHP

6. System Design

- Use Case Diagram
- Illustrate interactions between users, admins, and the system.
- Entity-Relationship (ER) Diagram

- Visualize the relationships between entities like Flights, Users, and Reservations.
- Data Flow Diagram (DFD)
- Depict how data flows between modules.

7. System Modules

- User Module
 - Registration and login.
 - Profile management
- Admin Module
 - Flight management.
 - User management.
- Reservation Module
 - Flight booking and cancellation.
 - Seat allocation
- Payment Module
 - Secure online payment integration.
 - Invoice and receipt generation

8. Database Design

Define tables such as:

- Users: UserID, Name, Email, Password
- Flights: FlightID, Source, Destination, Time
- Reservations: ReservationID, UserID, FlightID, SeatNumber

9. Implementation

Detail the technologies, tools, and frameworks used. Provide code snippets for key functionalities.

10. Testing

- Test Cases
 - List test scenarios for each module.
- Test Results
 - Provide the outcome of tests with screenshots or logs.

11. Security Measures

- Secure login (password hashing, 2FA).
- Data encryption for sensitive information.
- Role-based access control for admins and users.

12. Limitations

- Mention current system constraints, such as:
- Limited scalability.
- No multi-language support.

13. Future Enhancements

- Integration with third-party travel agencies
- Mobile app development.
- Real-time notifications for flight updates.

14. Conclusion

Summarize the project achievements and its impact on the airline industry, as delivered by Flight Center.

15. References

Books, articles, and websites referred to during development.